

# □ Cardiovascular Disease Risk Prediction

## Machine Learning Classification Study

Dataset: CDC BRFSS 2022 · 308,854 records · 18 features

Models: Logistic Regression · Decision Tree · Random Forest

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**308,854**

0.25

**18**

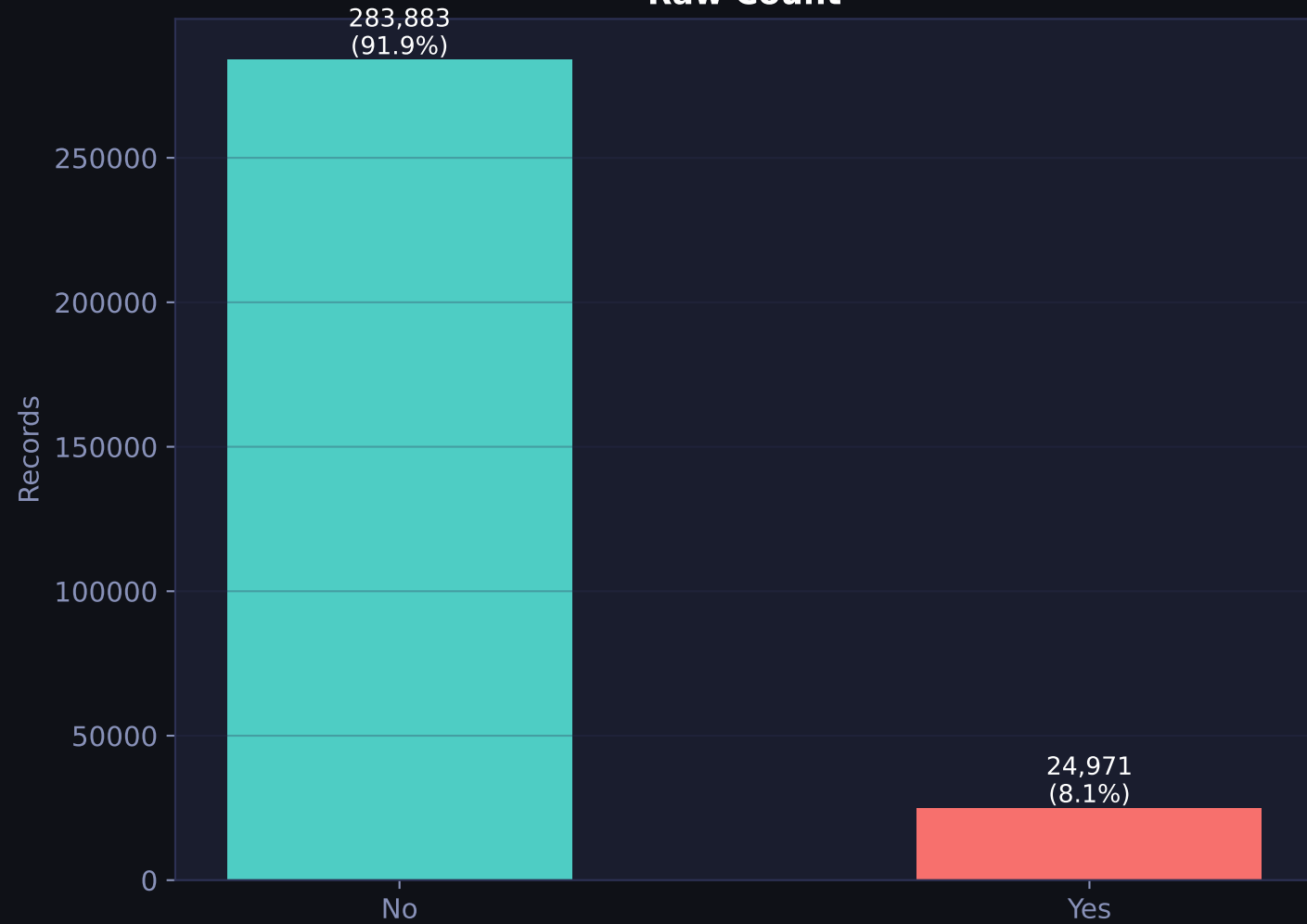
0.25

**4**

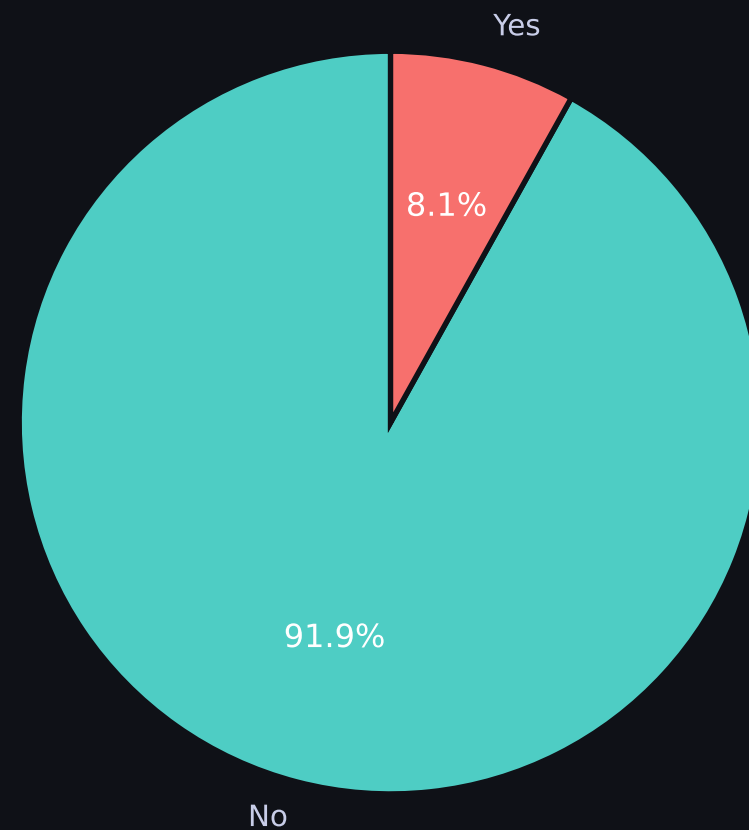
0.25

# Target Variable — Heart Disease Distribution

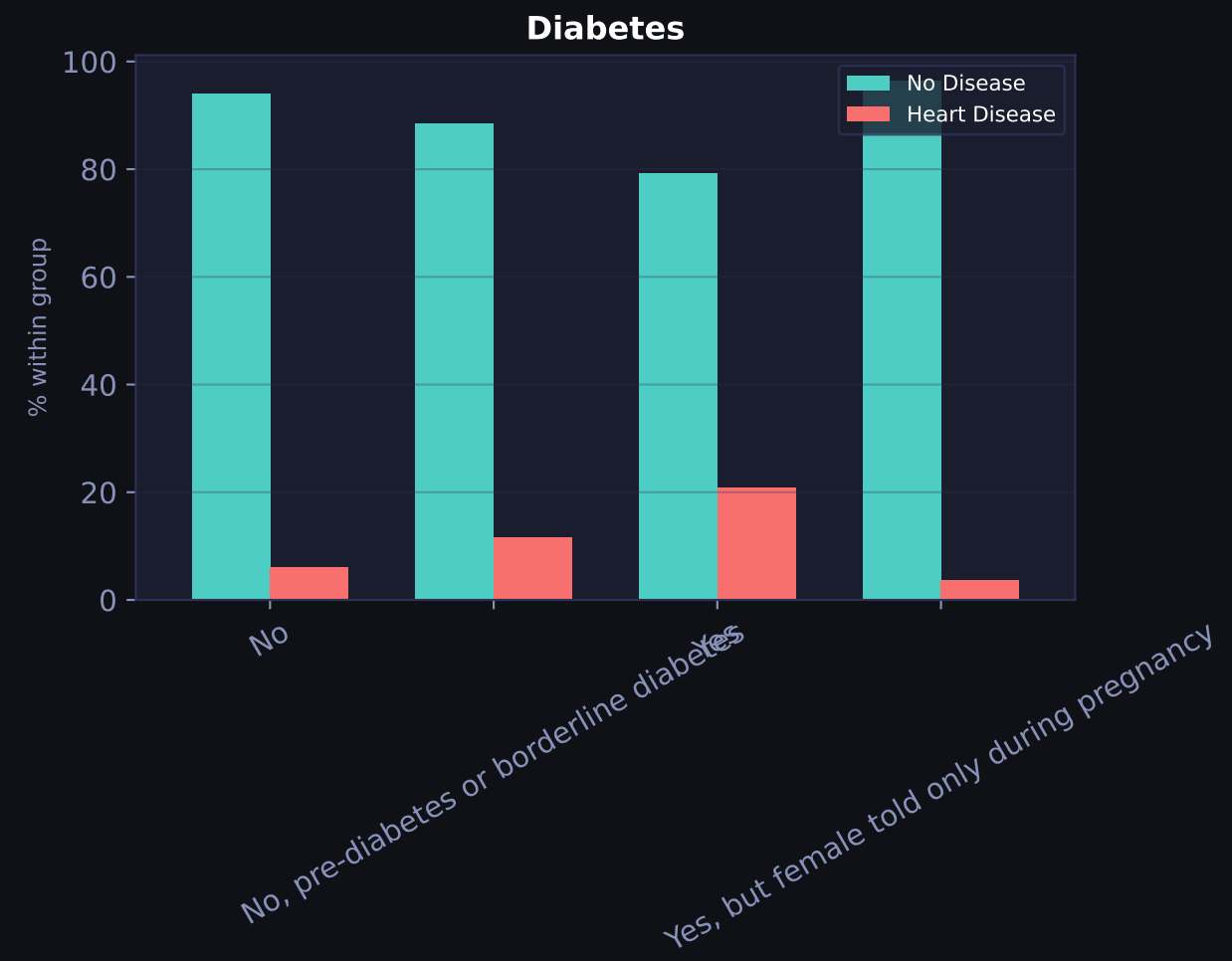
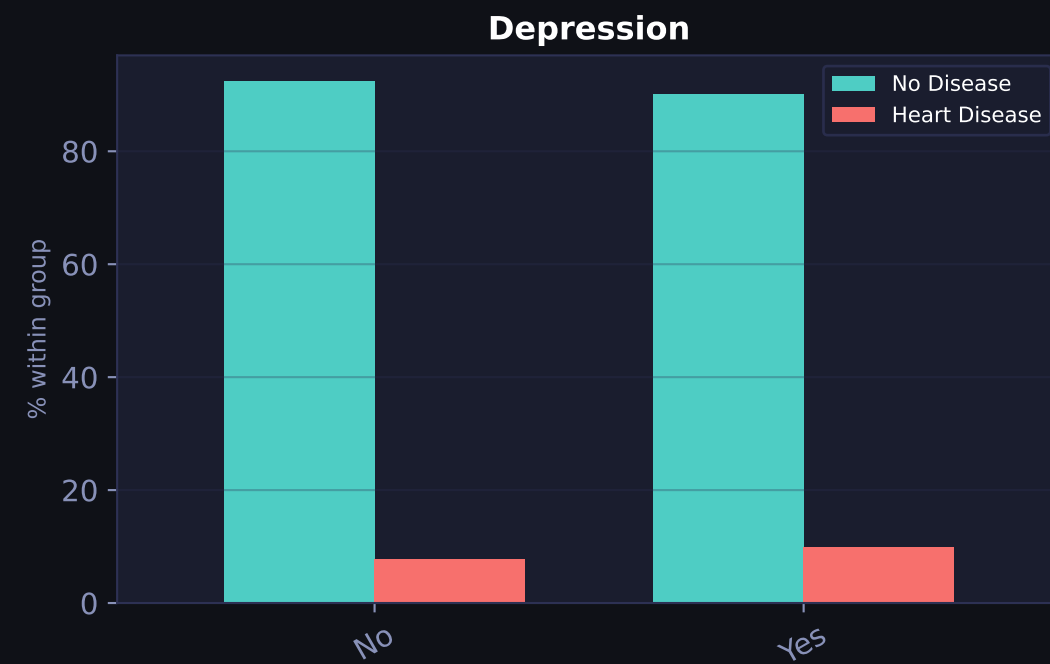
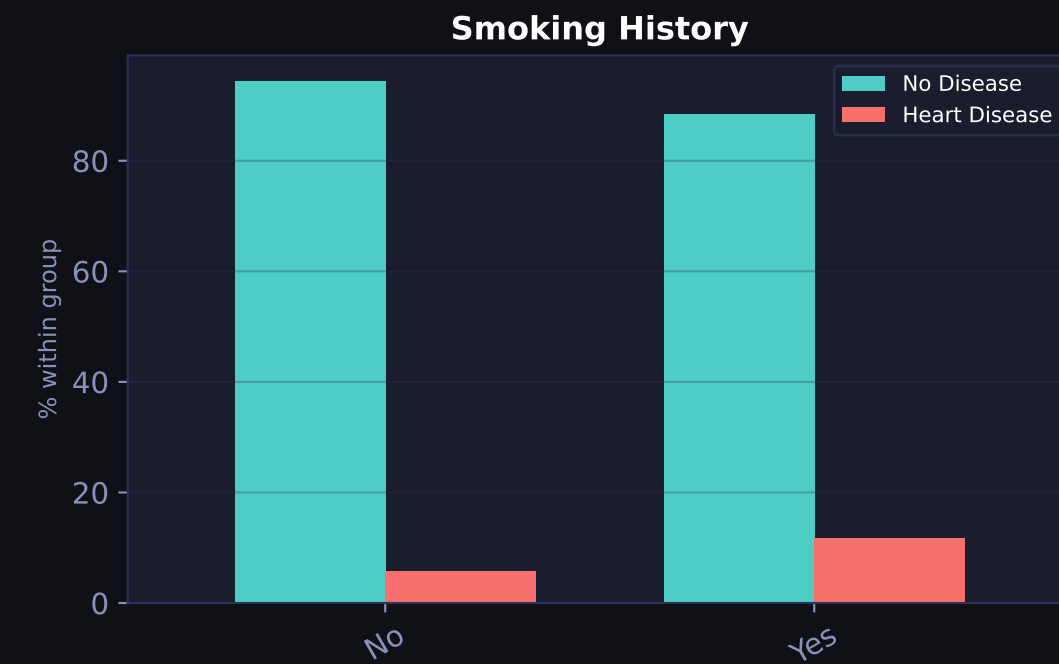
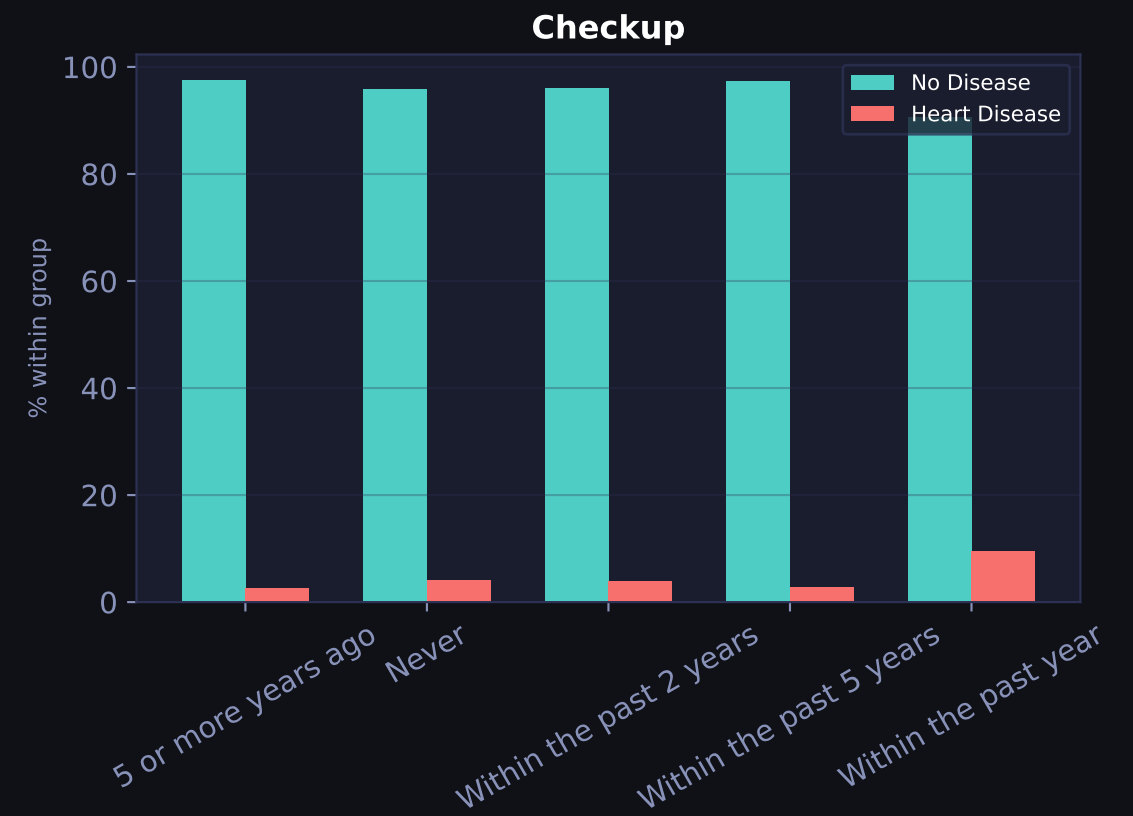
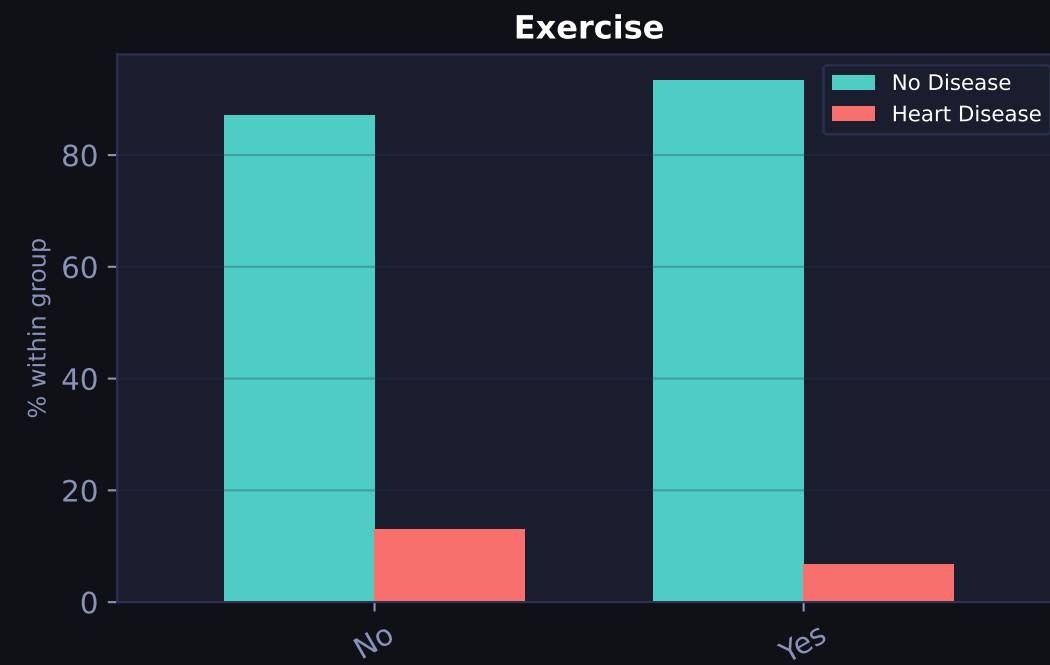
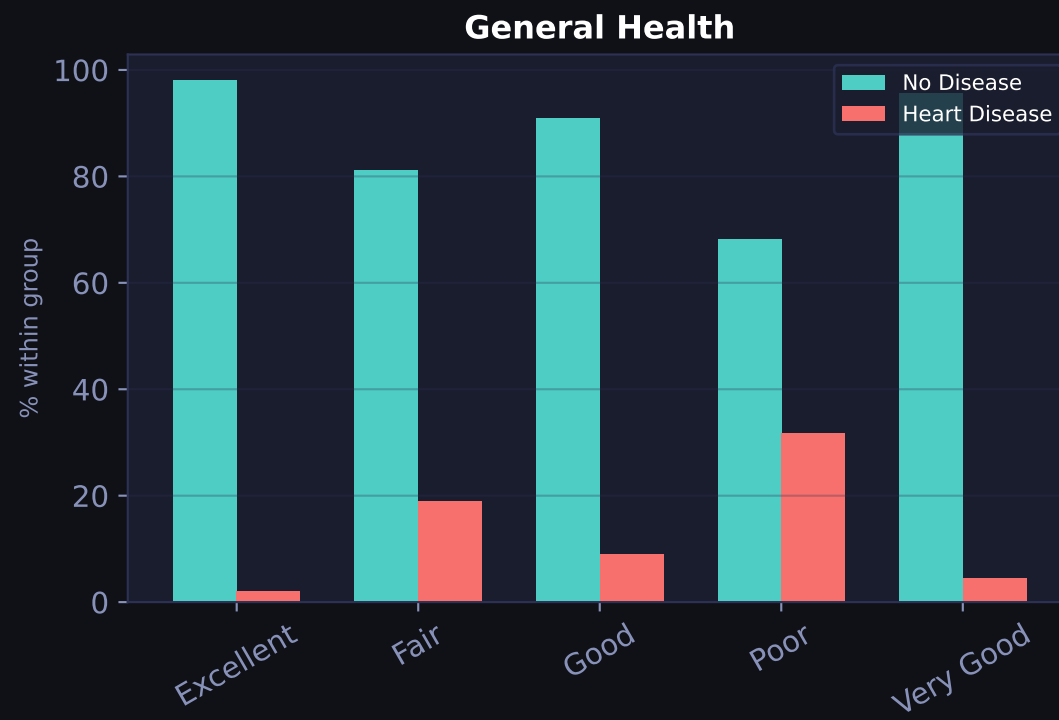
Raw Count



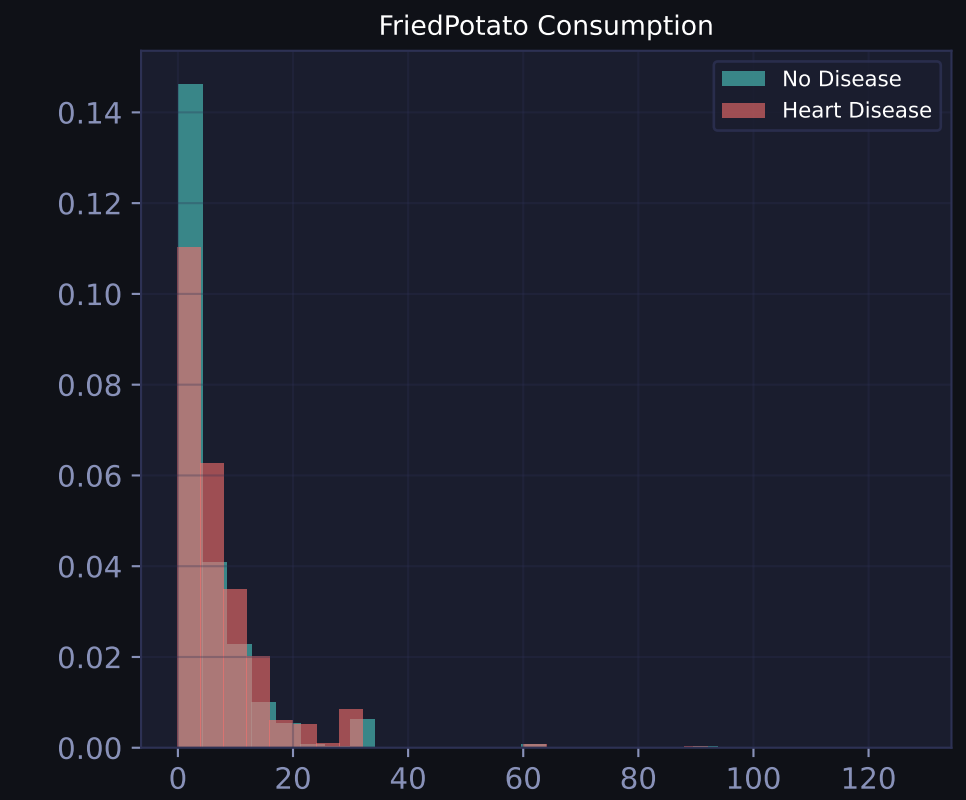
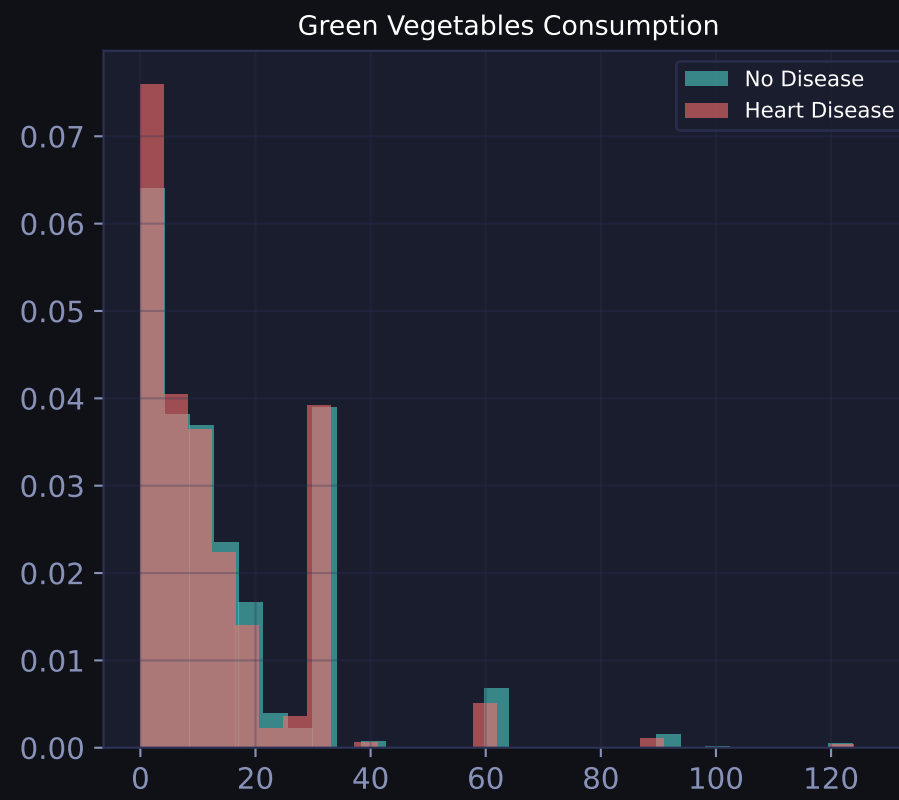
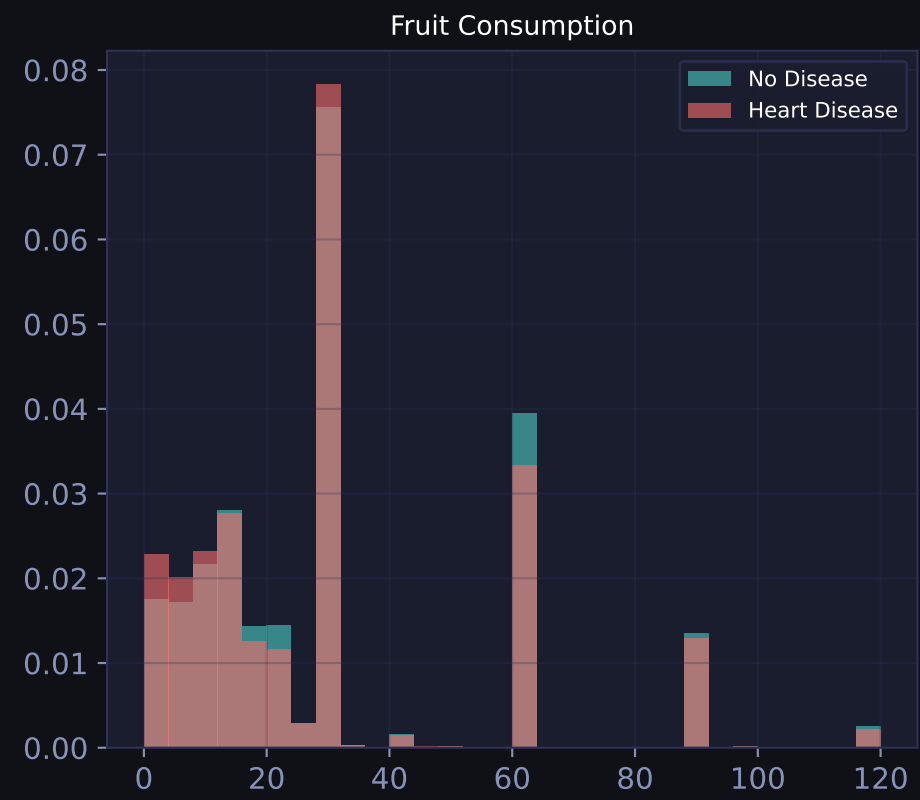
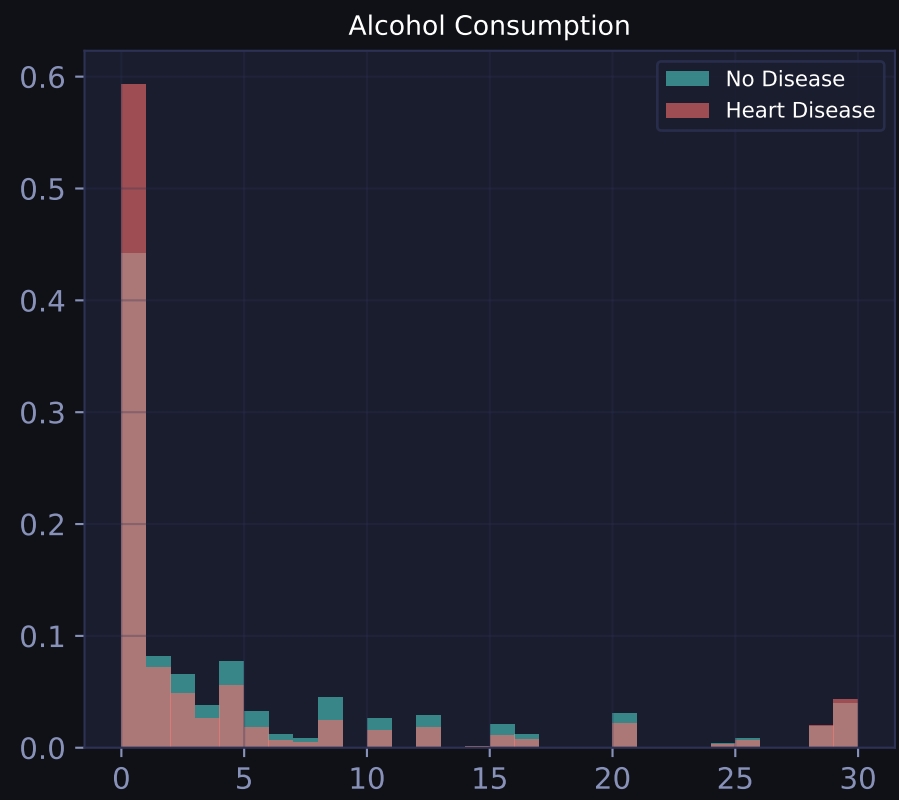
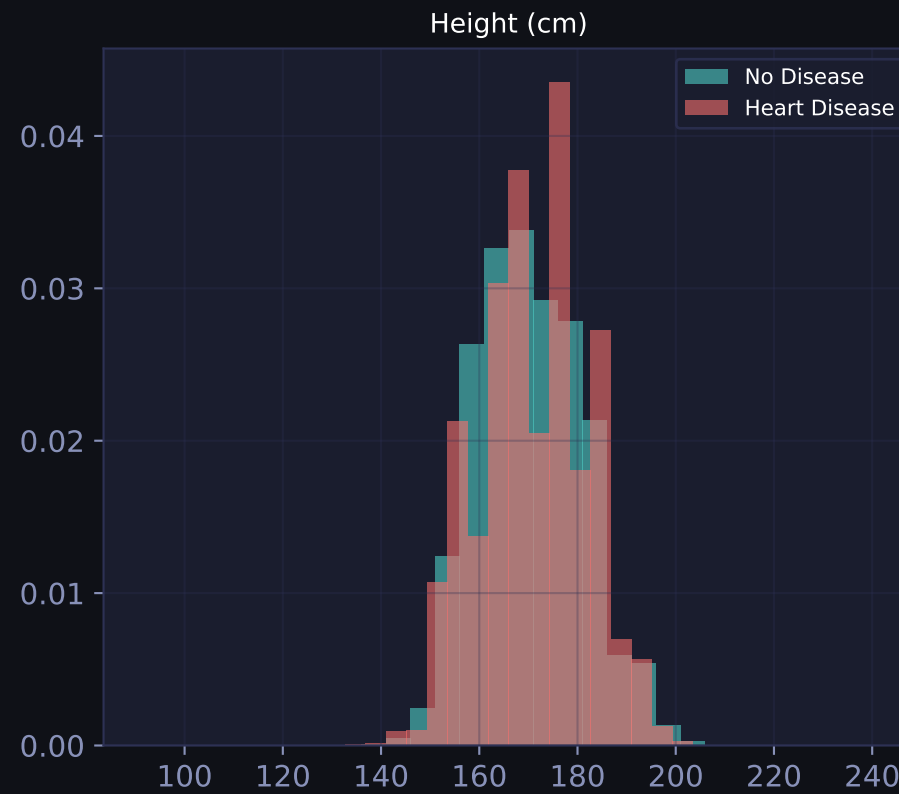
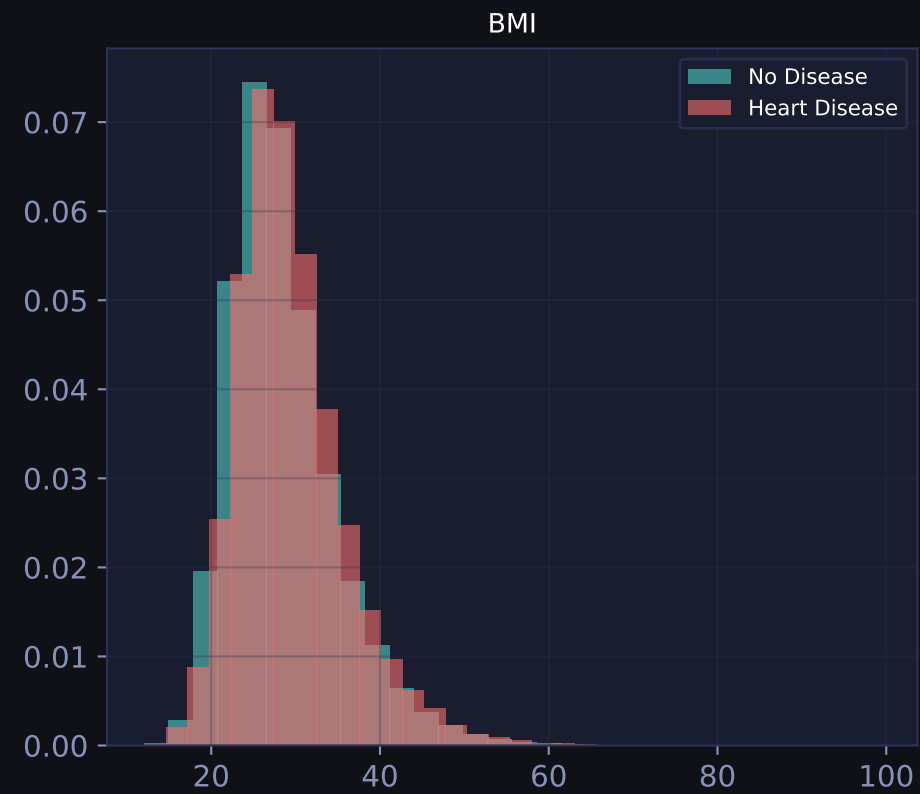
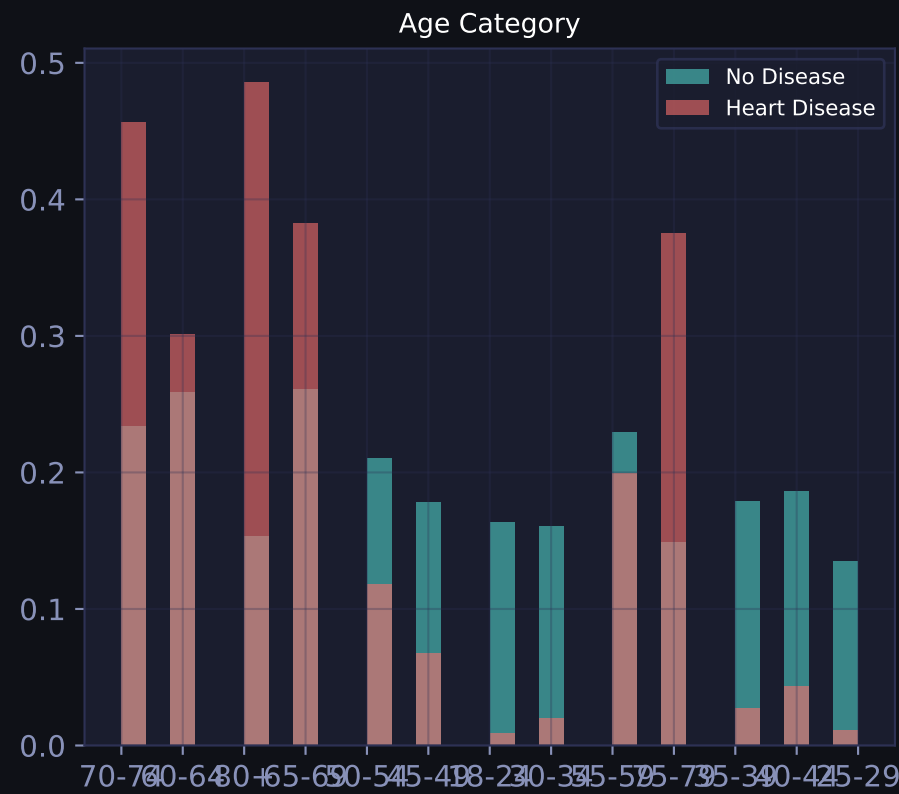
Class Proportion



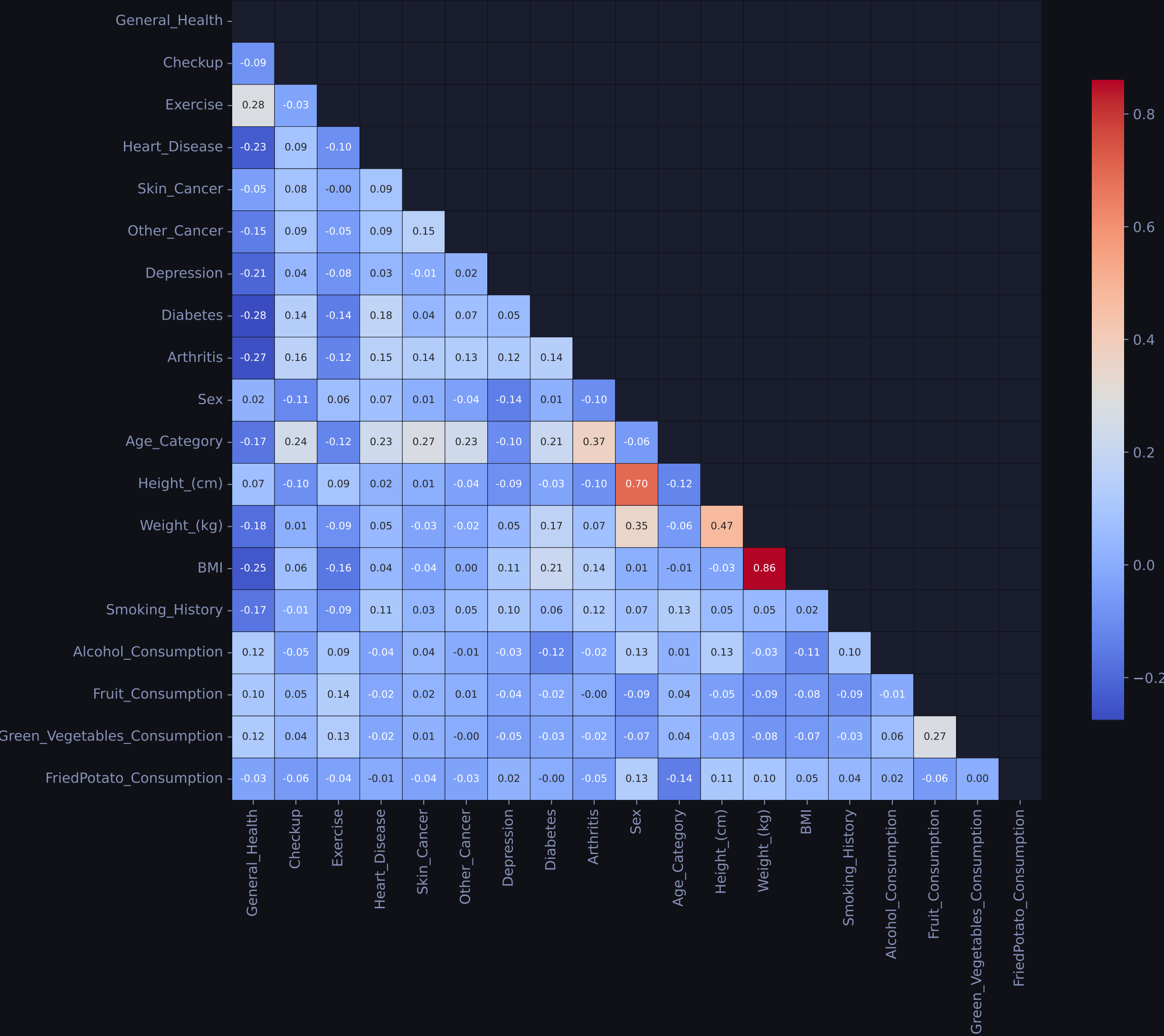
# Categorical Features vs Heart Disease Status



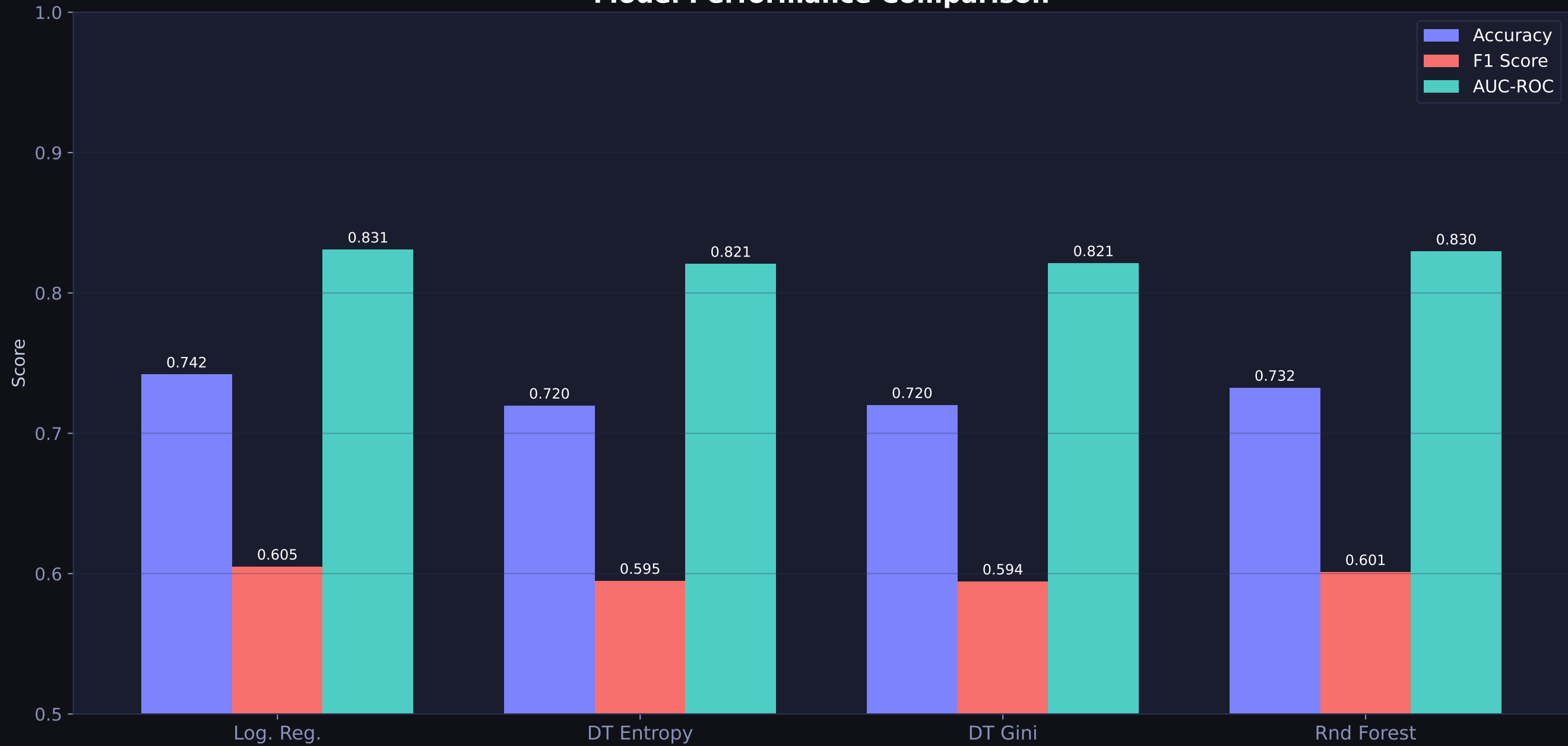
# Continuous Features by Heart Disease Status



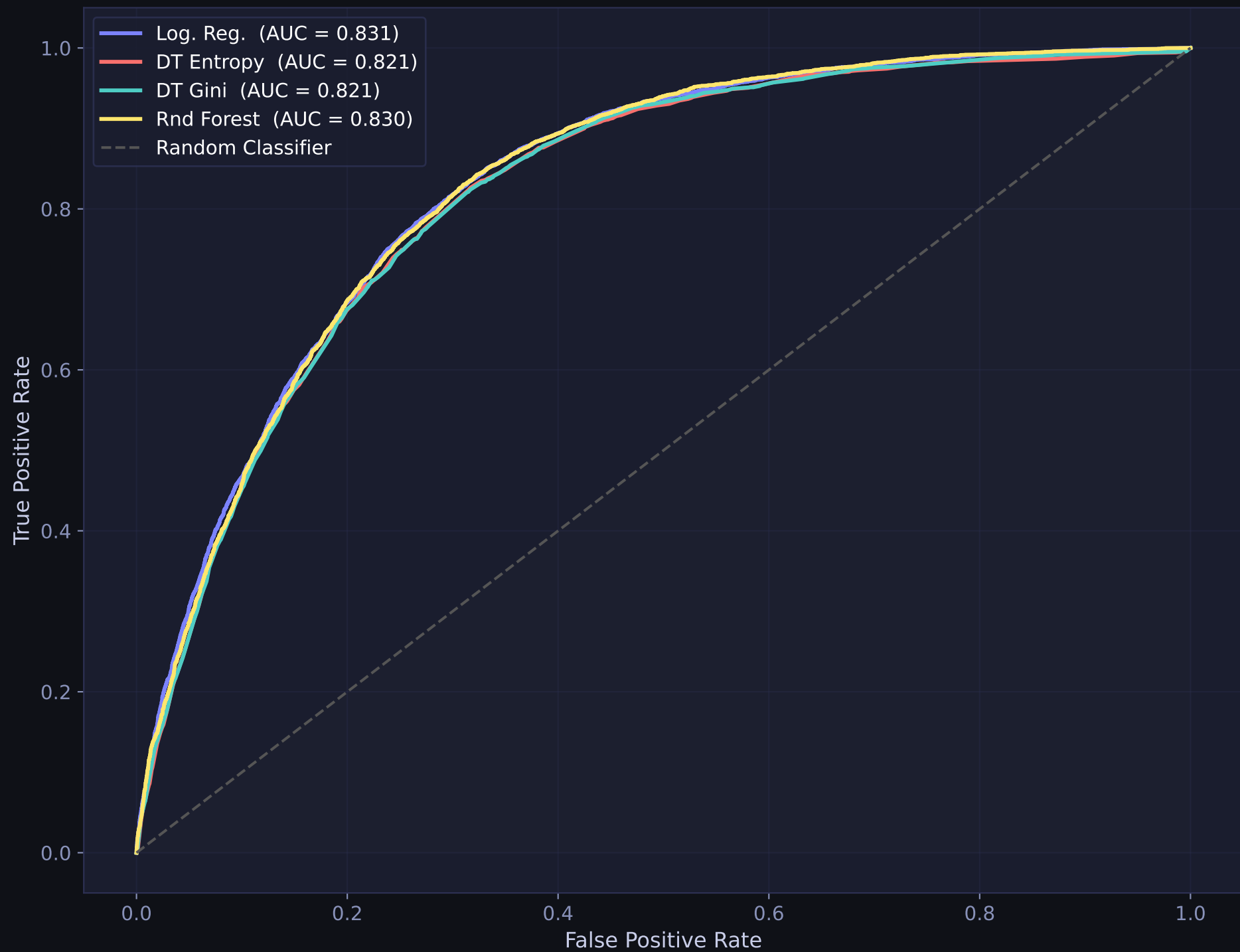
Feature Correlation Matrix



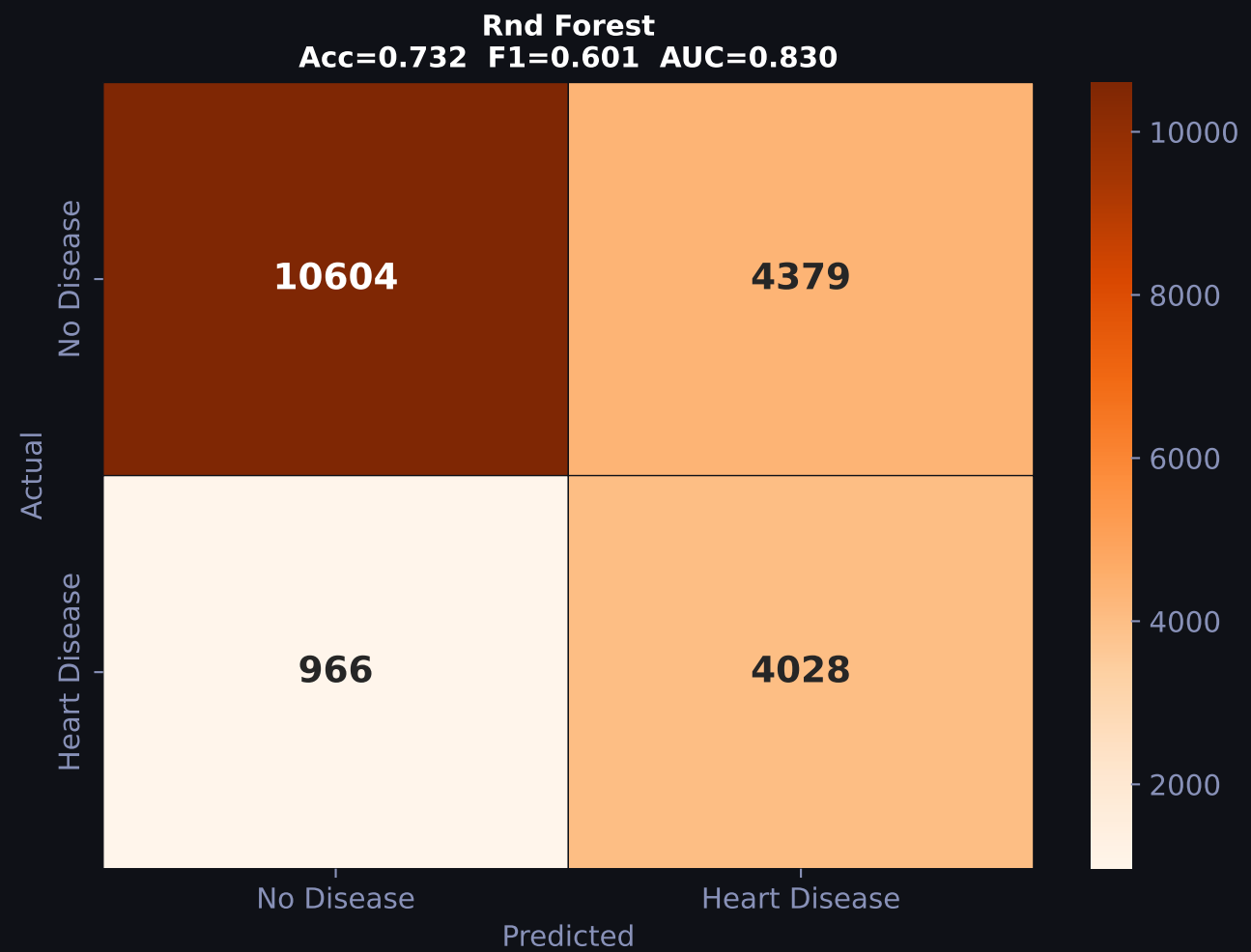
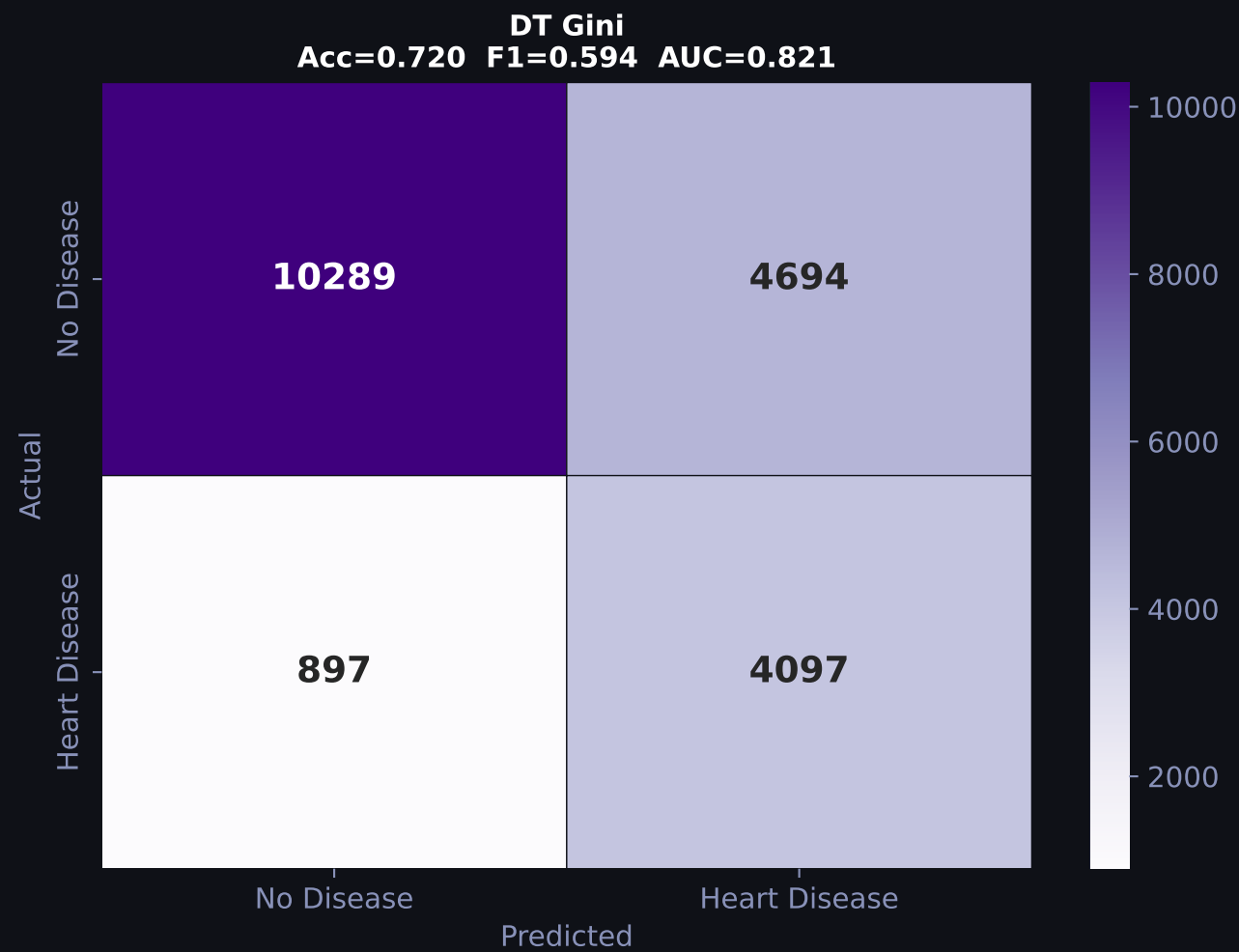
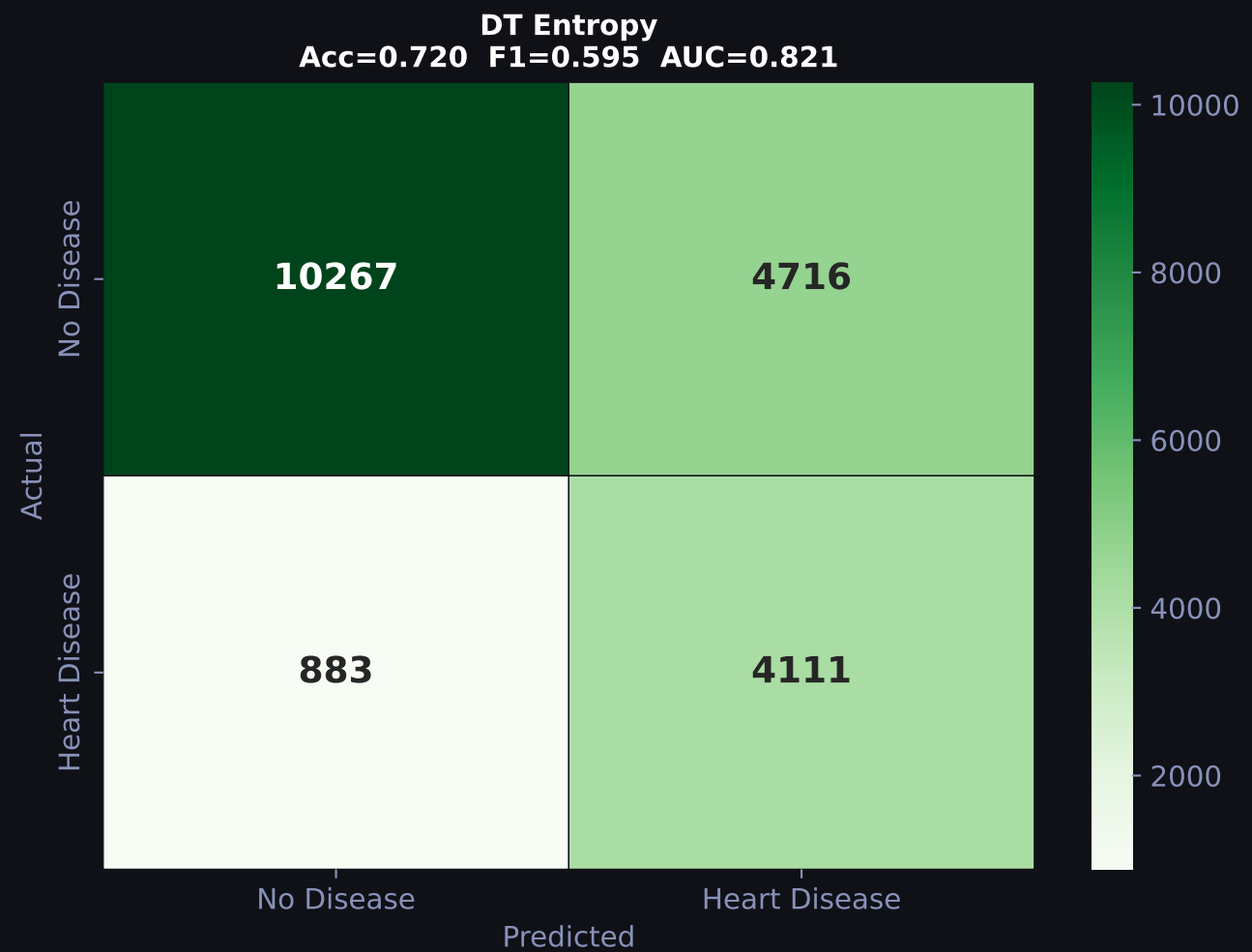
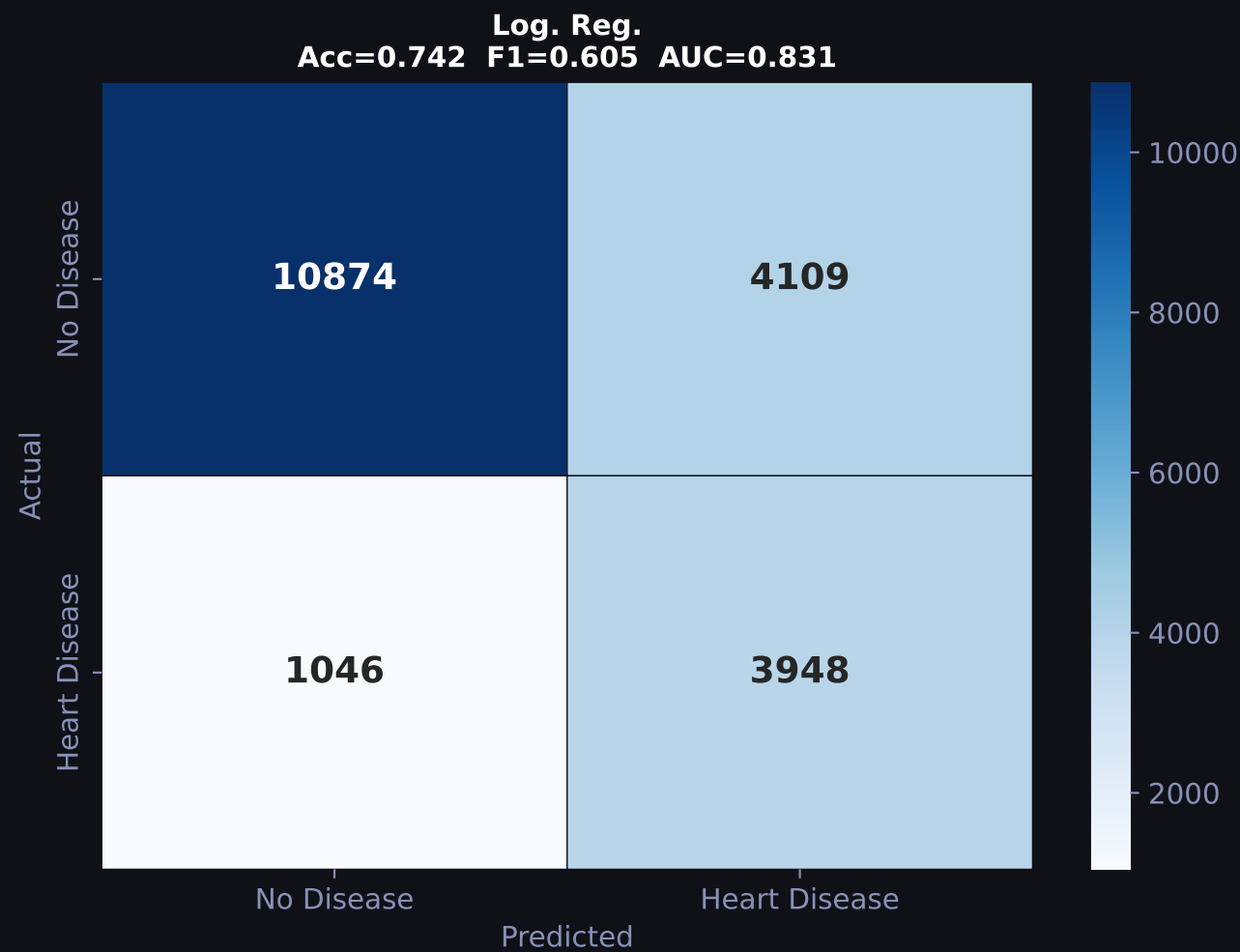
# Model Performance Comparison



# ROC Curves — All Models



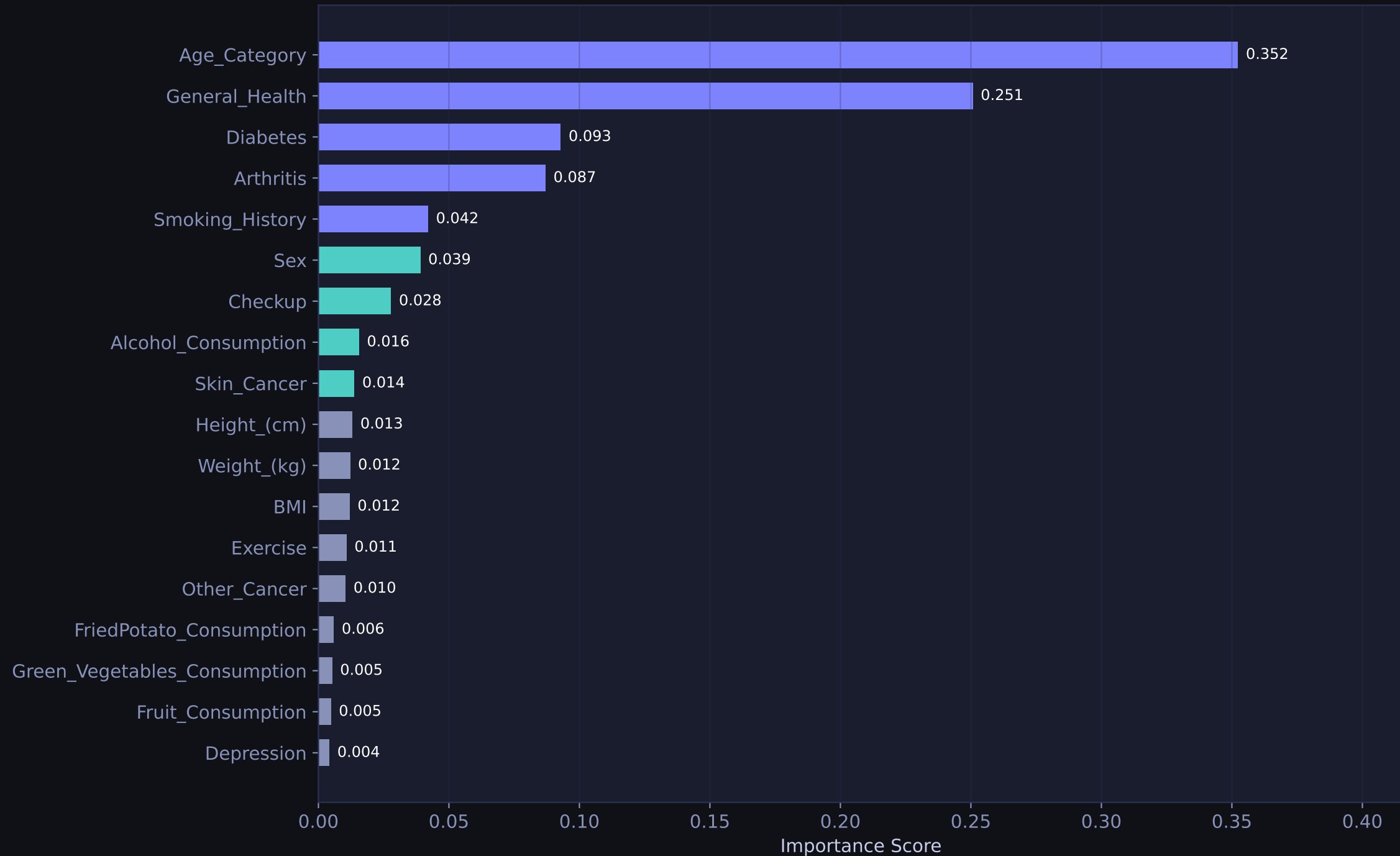
# Confusion Matrices — All Models





# Feature Importance — Random Forest

(violet = top quartile · teal = mid · grey = lower)



# Results Summary

| Model      | Accuracy | F1 Score | AUC-ROC | Best metric |
|------------|----------|----------|---------|-------------|
| Log. Reg.  | 0.7420   | 0.6050   | 0.8309  | 🏆 Best      |
| DT Entropy | 0.7197   | 0.5949   | 0.8208  |             |
| DT Gini    | 0.7201   | 0.5944   | 0.8212  |             |
| Rnd Forest | 0.7324   | 0.6011   | 0.8296  |             |

Key insight: AUC-ROC (~0.836) is the reliable metric here — not raw accuracy.

The dataset has 11.4:1 class imbalance; a naive "always No" model scores 91.9% accuracy without learning anything.

Top predictors: Age Category (36.7%) · General Health (24.5%) · Diabetes (9.4%) · Arthritis (8.8%)